

Shayan Poigai

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EDUCATION

Santa Clara University — B.S. in Computer Science

Expected December 2027

Major GPA: 4.0

Relevant Coursework: Advanced Algorithms, Data Structures & Algorithms, Advanced C++, Theory of Automata, Linear Algebra, Discrete Mathematics

WORK EXPERIENCE

Co-Founder & CTO/Lead Software Engineer

May 2026 – Present

Adorus (adorusjewels.com)

- Sole developer on a 4-person founding team, building the full-stack jewelry e-commerce platform end-to-end (pre-launch).
- Translating co-founder stakeholder feedback into data models, API contracts, and component architecture for the platform
- Deploying a responsive React.js/Node.js frontend backed by Java Spring Boot RESTful services and a PostgreSQL database, deployed on AWS EC2, with AWS S3 implemented for secure, cloud-based product image storage.

Software Engineering Intern

June 2026 – Present

HerbsPro

- Implementing backend features across the e-commerce stack.

Machine Learning Research Assistant

Dec 2025 – June 2026

Santa Clara University

- Co-authored a peer-reviewed paper as primary writer, accepted at HIBIBI 2026 (ASONAM-collocated), publishing with Springer.
- Built an NLP/data pipeline analyzing 5,000+ scraped posts from Twitter, Reddit, and YouTube to study public sentiment toward LLMs as a therapeutic tool, applying BERTopic, LDA, and NMF topic modeling.
- Engineered Pandas preprocessing to normalize unstructured text, reducing noise in high-dimensional social media datasets.
- Produced Matplotlib / Seaborn analytics on shifting attitudes toward AI mental-health support.

CS Teaching Assistant & Algorithms Grader

Sept 2025 – June 2026

Santa Clara University

- Mentored 40+ students in C++ (memory management, pointers, OOP) and debugged logic errors live using GDB
- Graded Theory of Algorithms coursework for 40+ students across 2 sections, evaluating runtime analysis, greedy, divide-and-conquer, and dynamic-programming solutions.

PROJECTS

Disease Tracker (AWS Hackathon)

Oct 2025

React, FastAPI, AWS Bedrock, DynamoDB

[Github](#)

- Built a FastAPI REST API serving real-time disease metrics and risk assessments with low-latency responses.
- Engineered an AWS Bedrock AI risk-scoring pipeline and designed DynamoDB schemas for scalable data ingestion.

Kuhn Poker vs Quantum

June 2026

Qiskit, IBM Quantum, FastAPI, React, Vercel, Render

[Github](#), [Live](#)

- Built a full-stack game where a 12 parameter, 6-qubit variational quantum circuit (superposition + entanglement) derives the opponent's mixed strategy, deployed on Vercel and Render.
- Ran the circuit on real IBM quantum hardware (e.g., `ibm_fez`, `ibm_mukesh`); on a 4,096-shot run, real-chip probabilities matched simulation within ~1.7 percentage points on average (max ~4), the gap attributable to quantum noise.

Social Network (C++ / Qt)

Sept 2025

C++, Qt, MVC, Graphs

[Github](#)

- Built a desktop social networking app with a Qt GUI and MVC architecture; modeled friendships with a custom graph and generated friend suggestions via BFS traversal, with file I/O for persistence.

Agentic RAG Pipeline

July 2025

Python, ChromaDB, Ollama, MCP, RAG

[Github](#)

- Built a modular RAG framework over ChromaDB and Ollama for semantic search across dynamic datasets.
- Implemented an MCP (Model Context Protocol) server exposing local data as tools for AI agents via JSON-RPC, designed for drop-in integration of new datasets.

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, SQL

Frameworks & Tools: React.js, Node.js, Spring Boot, FastAPI, Docker, Linux, Qt, Git, Vercel, CI/CD

Cloud & Data: AWS (EC2, Bedrock, Cognito, S3), PostgreSQL, DynamoDB, ChromaDB

AI / ML: RAG, MCP, NLP, Ollama, OpenWebUI, NumPy, Pandas, Matplotlib, Qiskit, topic modeling (BERTopic, LDA, NMF)